

Part: **4.4 – CRACK PROPAGATION ANALYSIS**

Chapter: **4440**

Title: **RESIDUAL FATIGUE LIFE**

Revision: **1.0**

Date: **03/05/2022**

1 INTRODUCTION

The most common method to perform the damage tolerance evaluation of a structural part is the crack propagation analysis. However, there are some cases in which it may be acceptable to apply the Residual Fatigue Life (RFL) approach to calculate the Inspection Interval.

The residual fatigue life approach may be used for:

- Structural elements designed as:
 - Multiple Load Path fail-safe redundant structures, for which each of the load paths is physically differentiated and the detectable condition (visual inspection) is a completely failed load path.
 - Waiting fail safe structures. This means that the component does not carry loads upon the failure of the primary load path.

After the primary load path failure, the loads are safely redistributed. It must be checked that the remaining load paths are capable to withstand fatigue loads and fail-safe limit loads.

- Structural elements not classified as Principal Structural Elements but whose failure may have an indirect impact on airworthiness. i.e. SSEs which are not PSEs.
- Structural elements for which applying fracture mechanics technics is not reliable enough. For example, redundant structural components such as bolts, shear pins, axles, door stops, cargo door hooks and hinges.

2 RESIDUAL FATIGUE LIFE METHOD

The main goal of the residual fatigue life assessment is the calculation on the remnant service life of the secondary load path, N_{RFL} , after the complete failure of the primer load path (N_{PF}). The calculated residual fatigue life N_{RFL} is then used to calculate the inspection interval I by applying the corresponding scatter factor.

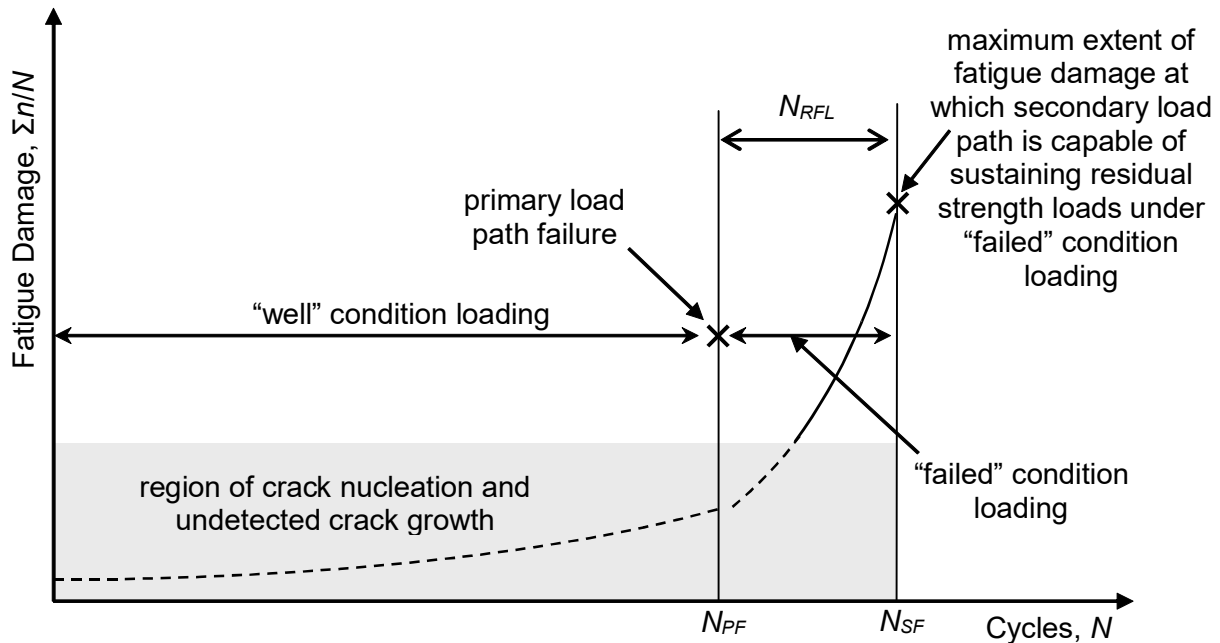


Figure 1. Residual fatigue approach to determining inspections intervals (from Ref. 1)

The methodology to calculate the fatigue damage is the same as the one described in chapter 4310 of the SMM (Crack initiation life calculation for metallic materials).

Primary load path

Regarding the primary load path, it must be checked that its failure is unlikely during the aircraft's design service goal (DSG):

$$N_{PF} \geq SF \cdot DSG$$

Where N_{PF} is the unfactored fatigue life of the primary load path and SF is the corresponding scatter factor.

Secondary load path – Residual Fatigue Life Assessment

The analysis of the secondary load path must account for the damage accumulated under well condition (D_{intact}) and after the primary load path failure (D_{RFL}). In the case of the “waiting fail safe” structures, the damage under well condition is null ($D_{intact} = 0$).

It is assumed that the failure of the part occurs when the total fatigue damage (D_{total}) exceeds a critical value of 1:

$$D_{total} = D_{intact} + D_{RFL} = 1$$

However, when substantiated by test evidences or fleet experience, other values different from the unity could be used.

Re-writing the above equations in terms of fatigue lives:

$$D_{intact} = \frac{DSG}{FL_{intact}} \quad D_{RFL} = \frac{N_{RFL}}{FL_{fail-safe}}$$

$$\frac{DSG}{FL_{intact}} + \frac{N_{RFL}}{FL_{fail-safe}} = 1$$

The Residual Fatigue Life (N_{RFL}) can be then calculated using the following equation:

$$N_{RFL} = FL_{fail-safe} \cdot \left(1 - \frac{DSG}{FL_{intact}}\right)$$

Where:

- $FL_{fail-safe}$ is the unfactored fatigue life of the secondary load path calculated under redistributed loads.
- DSG is the aircraft’s design service goal.
- FL_{intact} is the unfactored fatigue life of the secondary load path during the well condition. Conservatively, it has to be calculated for the DSG.

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
D	-	Damage
DSG	-	Design Service Goal
FL	-	Fatigue Life
II	-	Inspection Interval
N _{RFL}	-	Residual Fatigue Life
N _{PF}	-	Fatigue Life of the primary load path

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	X00RP0515001	1.0	Determination of inspection intervals by residual fatigue

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 03/05/2022	M.Carmen.Blanquer Darío Fernández	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEPAA)	Javier Gómez-Escalonilla 03/05/2022	Holger Hickethier 03/05/2022
Stress Methods (TEPAP-TL4)	Darío R. Fernández Villalba 03/05/2022	-

Table 3. Approval signatures table for last revision of this chapter